Term Project New York City Taxi Fare Prediction

2101-565

Data Analytics and Machine Learning
in Transportation Engineering

NYC Taxi Fare Prediction

- You are tasked with predicting the fare amount (inclusive of tolls) for a taxi ride in New York City given the pickup and dropoff locations.
- While you can get a basic estimate based on just the distance between the two points, this will result in an RMSE of $5-$8, depending on the model used.
- Your challenge is to do better than this using Machine Learning techniques.

Data Description (1)

Features

- **pickup_datetime** - timestamp value indicating when the taxi ride started.
- **pickup_longitude** - float for longitude coordinate of where the taxi ride started.
- **pickup_latitude** - float for latitude coordinate of where the taxi ride started.
- **dropoff_longitude** - float for longitude coordinate of where the taxi ride ended.
- **dropoff_latitude** - float for latitude coordinate of where the taxi ride ended.
- **passenger_count** - integer indicating the number of passengers in the taxi ride.

Data Description (2)

Target

- **fare_amount** - float dollar amount of the cost of the taxi ride. This value is only in the training set.

ID

- **key** - Unique string identifying each row in both the training and test sets. It should just be used as a unique ID field. Required in your submission CSV. Not necessarily needed in the training set but could be useful to simulate a 'submission file' while doing cross-validation within the training set.

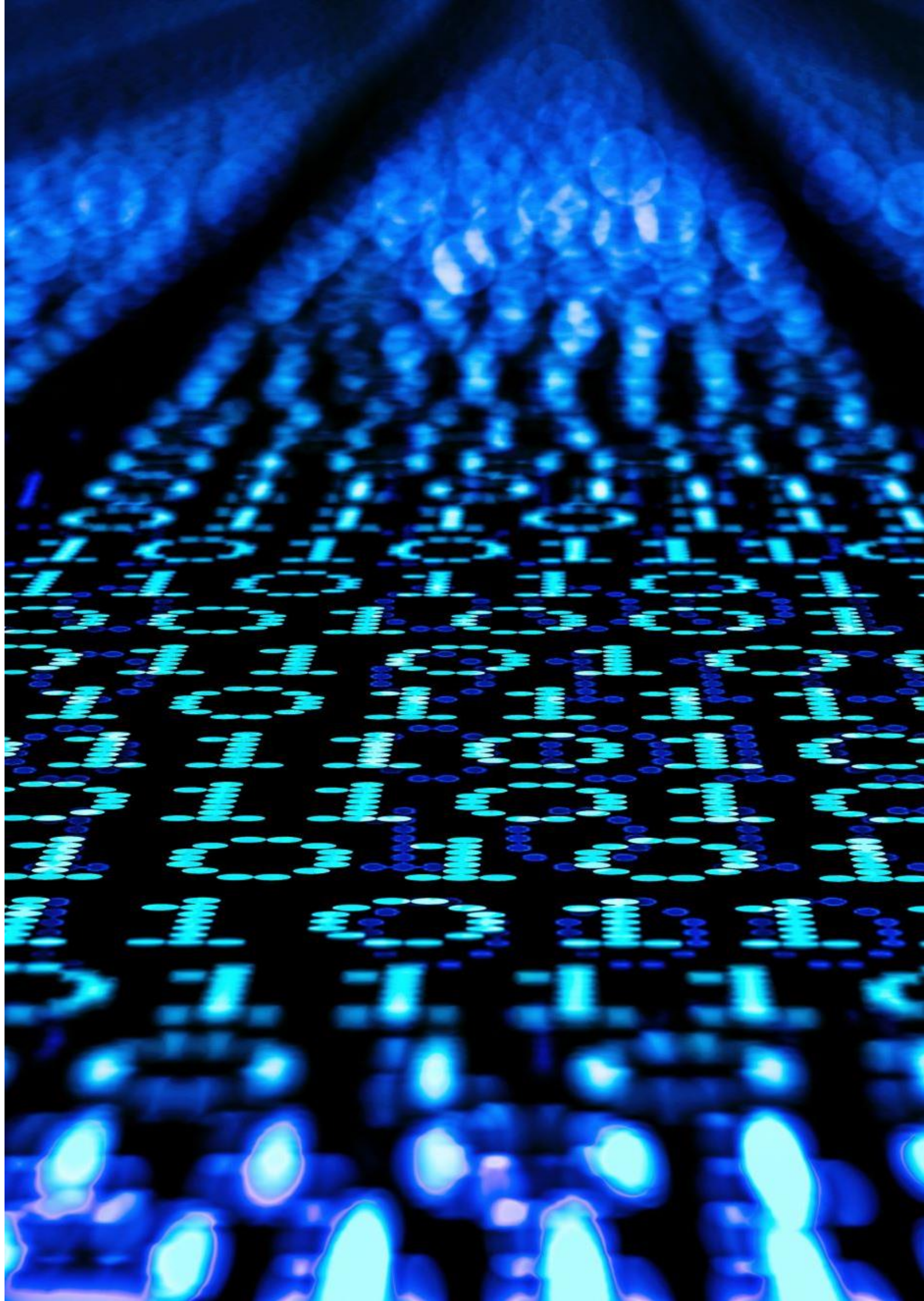
Evaluation

- The evaluation metric for this project is the root mean-squared error (RMSE).
- RMSE measures the difference between the predictions of a model, and the corresponding ground truth.
- A large RMSE is equivalent to a large average error, so smaller values of RMSE are better. One nice property of RMSE is that the error is given in the units being measured, so you can tell very directly how incorrect the model might be on unseen data.

Data Processing (1)

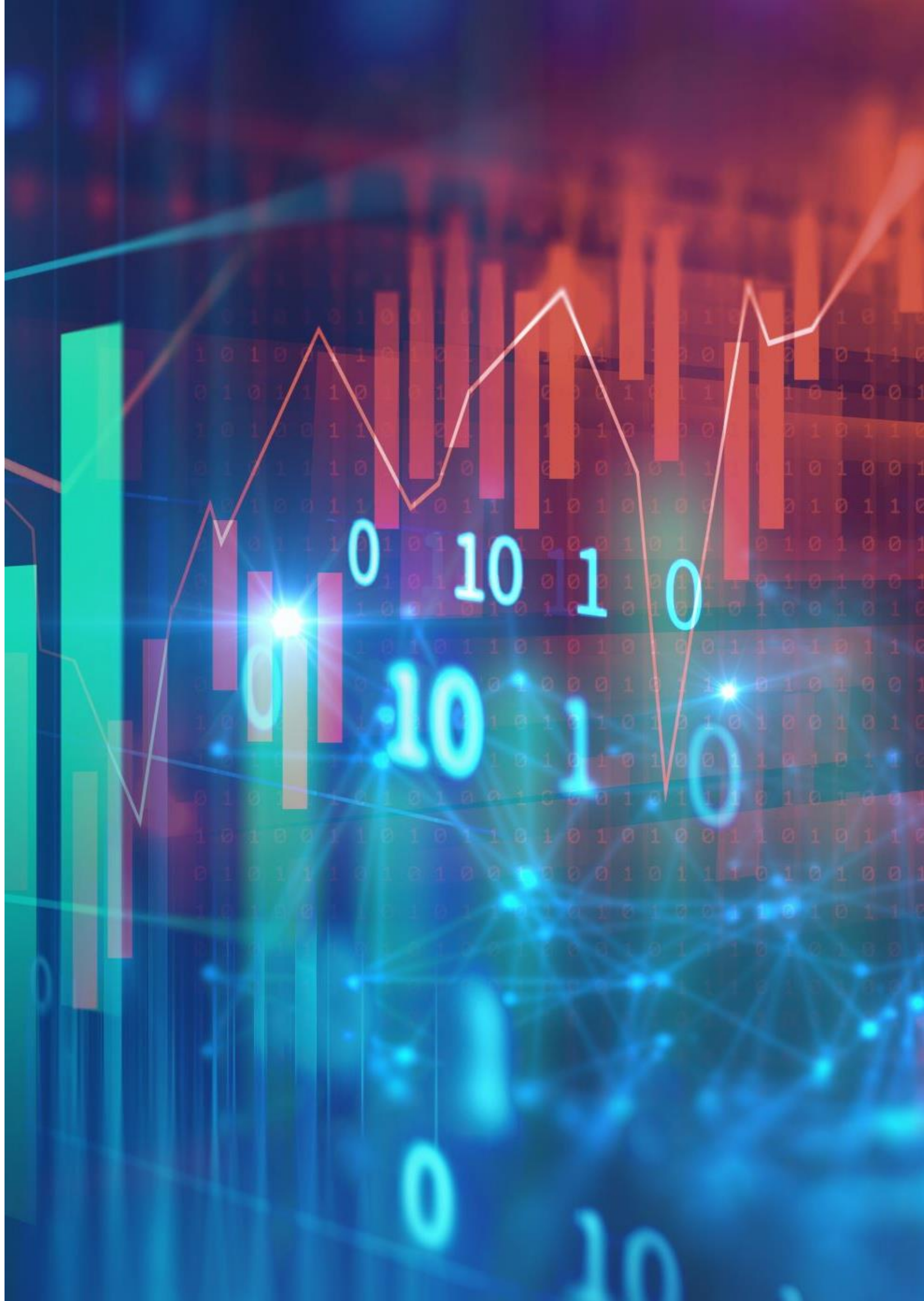
The following steps should be taken to clean and prepare the data for modeling:

- Removed outliers and invalid values such as negative or zero fares, distances, or durations.
- Imputed missing values with appropriate imputer depending on the feature.
- Created new features such as speed from existing features.
- Encoded categorical features with one-hot encoding.
- Scaled numerical features with standardization.



Data Processing (2)

- Dataset observations
- Data cleaning
- Feature Engineering
- Data visualization
- Fare Prediction
- Result Evaluation
- Conclusion



Algorithms To be used

Linear Regression
Polynomial Regression
Ridge/LASSO Regression
kNN Regressor
Support Vector Machines
Decision Tree Regression
Random Forest
XGBoost

Rules

- The external data are allowed.
- We ask that you respect the spirit of the competition and do not cheat.
- Hand-labeling is forbidden.